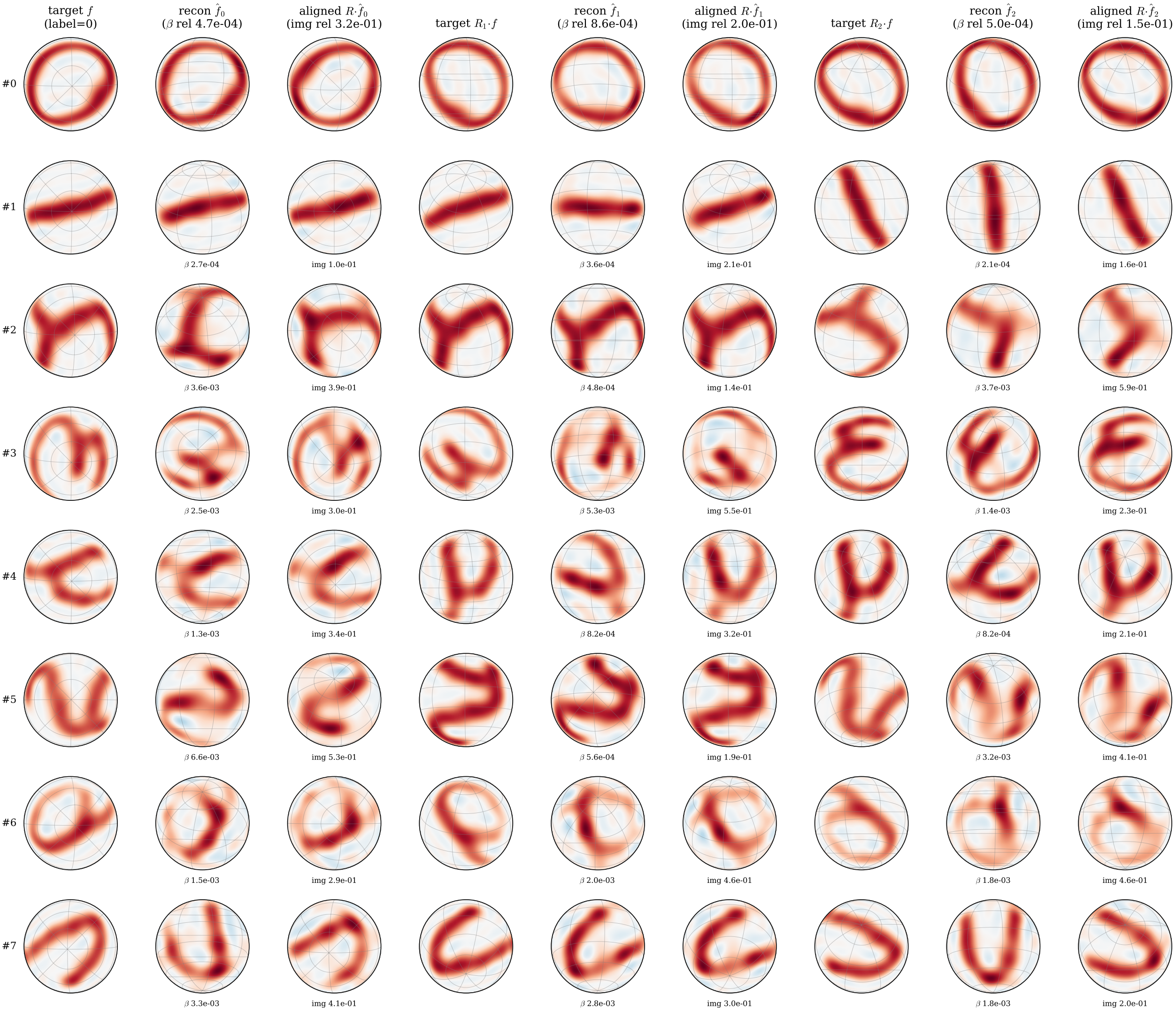
SO(3) bispectrum reconstruction.

Target $f_k = R_k \cdot f$ (left of each triplet), gradient-descent recon $\hat{f}_k$ from $\beta(f_k)$ (middle), and the recon rotated by the optimal $R \in SO(3)$ (right).

Each sphere is auto-centered on the signal's positive-mass centroid; aligned and target panels share the same view because they are the same signal.